

BEGUM ROKEYA UNIVERSITY, RANGPUR

DEPARTMENT OF STATISTICS

# STATISTICAL DATA ANALYSIS

*Comprehensive Laboratory Assignment Report*

## SUBMITTED BY

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# Problem 1

## Lung Disease Dataset Analysis using Python

### PROBLEM STATEMENT

Using the lung disease dataset, perform exploratory data analysis and fit a logistic regression model to identify key determinants of lung disease. Interpret your findings and discuss public health implications.

### A. IMPORT LIBRARIES AND DATASET

```
1 import pandas as pd
2 import numpy as np
3 from scipy.stats import chi2_contingency, fisher_exact
4 import statsmodels.formula.api as smf
5 from sklearn.metrics import roc_auc_score, confusion_matrix
6 from sklearn.metrics import accuracy_score, recall_score
7
8 df = pd.read_csv('lung_disease.csv')
9
10 print(df.head())
```

#### OUTPUT

|   | Age | Smoking | Pollution | Income | LungDisease |
|---|-----|---------|-----------|--------|-------------|
| 0 | 45  | Yes     | High      | Low    | Yes         |
| 1 | 32  | No      | Low       | High   | No          |
| 2 | 58  | Yes     | High      | Low    | Yes         |
| 3 | 27  | No      | Low       | High   | No          |
| 4 | 61  | Yes     | High      | Low    | Yes         |

### B. BASIC EDA QUESTIONS

#### i) Distribution of Age

```
1 print(df['Age'].describe())
2 print('Skewness =', df['Age'].skew())
```

**OUTPUT**

|       |            |
|-------|------------|
| count | 500.000000 |
| mean  | 43.904000  |
| std   | 14.604841  |
| min   | 20.000000  |
| 25%   | 30.750000  |
| 50%   | 45.000000  |
| 75%   | 56.000000  |
| max   | 69.000000  |

Skewness = 0.012

**Interpretation:** The average age is about 44 years. Age is approximately normally distributed with skewness very close to 0.

**ii) Proportion of Smokers vs Non-Smokers**

```
1 smoking_prop = df['Smoking'].value_counts(normalize=True) * 100
2 print(smoking_prop)
```

**OUTPUT**

|     |       |
|-----|-------|
| No  | 58.6% |
| Yes | 41.4% |

**Interpretation:** About 41.4% individuals are smokers, while 58.6% are non-smokers.

**iii) Income Group with Highest Frequency**

```
1 print(df['Income'].value_counts())
```

**OUTPUT**

|        |     |
|--------|-----|
| Low    | 204 |
| Medium | 197 |
| High   | 99  |

**Interpretation:** The Low income group has the highest frequency (204 individuals).

**iv) Percentage Exposed to High Pollution**

```
1 high_pollution = (df['Pollution'] == 'High').mean() * 100
2 print(high_pollution)
```

**OUTPUT**

70.4%

## v) Overall Prevalence of Lung Disease

```
1 prevalence = (df['LungDisease'] == 'Yes').mean() * 100
2 print(prevalence)
```

### OUTPUT

52.8%

## C. ASSOCIATION AND CROSSTAB - SMOKING VS LUNG DISEASE

```
1 crosstab_smoking = pd.crosstab(df['Smoking'], df['LungDisease'])
2 print(crosstab_smoking)
```

### OUTPUT

| LungDisease | No  | Yes |
|-------------|-----|-----|
| Smoking     |     |     |
| No          | 184 | 109 |
| Yes         | 52  | 155 |

## D. CHI-SQUARE TEST FOR ASSOCIATION

```
1 chi2, p, dof, expected = chi2_contingency(crosstab_smoking)
2 print('Chi-square =', chi2)
3 print('p-value =', p)
```

### OUTPUT

Chi-square = 67.59  
p-value = 2.01e-16

**Interpretation:** Smoking and Lung Disease are significantly associated (p < 0.05).

## E. ODDS RATIO FOR SMOKING

```
1 oddsratio, p = fisher_exact([[155, 52], [109, 184]])
2 print('Odds Ratio =', oddsratio)
```

### OUTPUT

Odds Ratio = 5.03

**Interpretation:** Smokers have approximately 5 times higher odds of developing lung disease.

## F. LOGISTIC REGRESSION MODEL

```
1 df2 = df.copy()
2 df2['Smoking'] = df2['Smoking'].map({'Yes':1, 'No':0})
3 df2['Pollution'] = df2['Pollution'].map({'High':1, 'Low':0})
4 df2['LungDisease'] = df2['LungDisease'].map({'Yes':1, 'No':0})
5
6 model = smf.logit('LungDisease ~ Smoking + Age + Pollution + C(
    Income)', data=df2).fit()
7 print(model.summary())
```

### OUTPUT

| Variable       | Coef  | p-value |
|----------------|-------|---------|
| Smoking        | 1.702 | <0.001  |
| Age            | 0.040 | <0.001  |
| Pollution      | 1.303 | <0.001  |
| Income(Low)    | 0.440 | 0.123   |
| Income(Medium) | 0.220 | 0.440   |

**Interpretation:** Smoking, Age, and Pollution are statistically significant predictors. Income is not statistically significant.

## G. ROC CURVE AND AUC

```
1 pred_prob = model.predict(df2)
2 auc = roc_auc_score(df2['LungDisease'], pred_prob)
3 print('AUC =', auc)
```

### OUTPUT

AUC = 0.787

## H. CONFUSION MATRIX AND PERFORMANCE METRICS

```
1 pred_class = (pred_prob > 0.5).astype(int)
2 cm = confusion_matrix(df2['LungDisease'], pred_class)
3 accuracy = accuracy_score(df2['LungDisease'], pred_class)
4 sensitivity = recall_score(df2['LungDisease'], pred_class)
5 specificity = 156 / (156 + 80)
6
7 print('Confusion Matrix:\n', cm)
8 print('Accuracy =', accuracy)
9 print('Sensitivity =', sensitivity)
10 print('Specificity =', specificity)
```

OUTPUT

Confusion Matrix:  
[[156 80]  
[ 64 200]]  
Accuracy = 0.712  
Sensitivity = 0.758  
Specificity = 0.661

I. FINAL CONCLUSION

- Smoking is the strongest predictor of lung disease (OR = 5.03)
- High pollution exposure significantly increases lung disease risk
- Older individuals are more vulnerable to lung disease
- Income level is not statistically significant
- The model achieves 71.2% accuracy with good sensitivity (75.8%)
- Public health policies should focus on smoking prevention and pollution control

## Problem 2

### One-Way ANOVA: Exercise Programs and Weight Loss

#### PROBLEM STATEMENT

Determine if three different exercise programs (A, B, C) impact weight loss differently. Recruit 90 people, randomly assign 30 to each program. Response variable: weight loss (pounds).

#### PYTHON CODE AND RESULTS

```
1 import numpy as np
2 import pandas as pd
3 from scipy.stats import f_oneway, shapiro, levene
4 from statsmodels.formula.api import ols
5 import statsmodels.api as sm
6 from statsmodels.stats.multicomp import pairwise_tukeyhsd
7
8 np.random.seed(42)
9 program_A = np.random.normal(5, 1.5, 30)
10 program_B = np.random.normal(7, 1.5, 30)
11 program_C = np.random.normal(5.5, 1.5, 30)
12
13 df = pd.DataFrame({
14     'WeightLoss': np.concatenate([program_A, program_B, program_C
15     ]),
16     'Program': ['A']*30 + ['B']*30 + ['C']*30
17 })
18
19 model = ols('WeightLoss ~ Program', data=df).fit()
20 anova_table = sm.stats.anova_lm(model, typ=2)
21 print(anova_table)
```

#### OUTPUT

|          | sum_sq     | df   | F         | PR(>F)       |
|----------|------------|------|-----------|--------------|
| Program  | 67.416974  | 2.0  | 16.890178 | 6.342321e-07 |
| Residual | 173.629808 | 87.0 | NaN       | NaN          |

#### ASSUMPTION CHECKING

```
1 shapiro_p = stats.shapiro(model.resid)[1]
2 levene_p = stats.levene(df['WeightLoss'][df['Program']=='A'],
3                           df['WeightLoss'][df['Program']=='B'],
4                           df['WeightLoss'][df['Program']=='C'])[1]
5 print(f"Shapiro-Wilk p-value: {shapiro_p:.4f}")
6 print(f"Levene's p-value: {levene_p:.4f}")
```

OUTPUT

Shapiro-Wilk p-value: 0.8948  
Levene's p-value: 0.8627

**Interpretation:** Both assumptions (normality and homogeneity of variance) are satisfied ( $p \geq 0.05$ ).

POST-HOC ANALYSIS (TUKEY HSD)

OUTPUT

| Multiple Comparison of Means - Tukey HSD, FWER=0.05 |        |          |        |         |         |        |
|-----------------------------------------------------|--------|----------|--------|---------|---------|--------|
| group1                                              | group2 | meandiff | p-adj  | lower   | upper   | reject |
| A                                                   | B      | 2.1005   | 0.000  | 1.2307  | 2.9702  | True   |
| A                                                   | C      | 0.8015   | 0.0772 | -0.0682 | 1.6713  | False  |
| B                                                   | C      | -1.2989  | 0.0017 | -2.1687 | -0.4292 | True   |

INTERPRETATION:

- Program B shows significantly higher weight loss than Program A ( $p \leq 0.001$ )
- Program B shows significantly higher weight loss than Program C ( $p \leq 0.01$ )
- No significant difference between Programs A and C

Problem 3

Chi-Square Analysis on Nigeria Child Anemia Dataset

DATASET DESCRIPTION

Source: 2018 Nigeria Demographic and Health Surveys (NDHS). Investigates factors affecting anemia levels among children aged 0-59 months.

PYTHON CODE AND RESULTS

```
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 from scipy.stats import chi2_contingency
6
7 df = pd.read_csv('children_anemia.csv')
8 clean_df = df.dropna(subset=['Wealth index combined', 'Anemia
   level'])
9 contingency_table = pd.crosstab(clean_df['Wealth index combined'],
10                                clean_df['Anemia level'])
11 chi2, p_val, dof, expected = chi2_contingency(contingency_table)
12
13 print(f"Chi-Square Statistic: {chi2:.2f}")
14 print(f"P-value: {p_val:.4e}")
```

OUTPUT

Chi-Square Statistic: 203.15  
P-value: 7.2798e-37  
Degrees of Freedom: 12

CONTINGENCY TABLE (PERCENTAGES BY WEALTH INDEX)

OUTPUT

| Anemia level          | Mild  | Moderate | Not anemic | Severe |
|-----------------------|-------|----------|------------|--------|
| Wealth index combined |       |          |            |        |
| Middle                | 28.99 | 30.16    | 39.20      | 1.66   |
| Poorer                | 27.44 | 32.52    | 38.12      | 1.92   |
| Poorest               | 28.39 | 35.66    | 33.89      | 2.06   |
| Richer                | 26.84 | 28.74    | 42.63      | 1.79   |
| Richest               | 24.29 | 22.17    | 52.29      | 1.25   |

## HYPOTHESES

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$H_0$  : Wealth index and child anemia level are independent

$H_1$  : Wealth index and child anemia level are associated

## INTERPRETATION

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Since  $p < 0.05$ , we reject the null hypothesis. There is a statistically significant association between wealth index and child anemia level.

## KEY FINDINGS

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- The "Poorest" group is significantly over-represented in severe anemia cases
- The "Richest" group shows the lowest anemia rates (only 1.25% severe)
- 52.29% of children in the richest families are not anemic vs only 33.89% in poorest families

## PUBLIC HEALTH IMPLICATIONS

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- Target poverty reduction interventions
- Nutritional awareness campaigns for low-income families
- Improved healthcare access in poor communities

Problem 4

Fisher’s Exact Test: Drug Treatments and Disease Outcome

PROBLEM STATEMENT

Suppose, there are three drug treatments (drug A, drug B, and drug C) with the outcome of a disease or no disease. We need to test if there is an association between drug treatments and disease outcomes. Perform Fisher’s exact test for the above contingency table, post-hoc test for Fisher’s exact test and interpret your findings.

GIVEN CONTINGENCY TABLE

| Treatment<br>Success Rate | No Disease | Disease |
|---------------------------|------------|---------|
| Drug A<br>80%             | 40         | 10      |
| Drug B<br>20%             | 10         | 40      |
| Drug C<br>50%             | 25         | 25      |

PYTHON CODE AND RESULTS

```
1 import numpy as np
2 from scipy.stats import chi2_contingency, fisher_exact
3 from statsmodels.stats.multitest import multipletests
4
5 table = np.array([[40, 10], [10, 40], [25, 25]])
6 chi2_stat, p_global, dof, expected = chi2_contingency(table)
7 print(f"Global P-value: {p_global:.4e}")
8
9 pairs = [(0,1), (0,2), (1,2)]
10 p_values = []
11 for i, j in pairs:
12     _, p_pair = fisher_exact(table[[i, j]])
13     p_values.append(p_pair)
14
15 reject, p_adjusted, _, _ = multipletests(p_values, method='
    bonferroni')
```

OUTPUT

Global P-value: 1.5230e-08  
Drug A vs Drug B: Adjusted P = 0.0000 (Significant: True)  
Drug A vs Drug C: Adjusted P = 0.0092 (Significant: True)  
Drug B vs Drug C: Adjusted P = 0.0092 (Significant: True)

INTERPRETATION

- **Drug A (80% success):** Most effective treatment
- **Drug B (20% success):** Least effective treatment
- **Drug C (50% success):** Intermediate effectiveness
- All pairwise differences are statistically significant after Bonferroni correction

## Problem 5

### Logistic Regression GLM - Graduate School Admission

#### PROBLEM STATEMENT

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable. The data set taken from <https://stats.idre.ucla.edu/stat/data/binary.csv> and fit logistic generalized linear models (GLMs) to identify the effect admission into graduate school. Interpret the result.

#### PYTHON CODE

```
1 import pandas as pd
2 import statsmodels.formula.api as smf
3
4 df = pd.read_csv('binary.csv')
5 model = smf.logit('admit ~ gre + gpa + C(rank)', data=df).fit()
6 print(model.summary())
```

#### MODEL SUMMARY

##### OUTPUT

|               | coef    | std err | z      | P> z  |
|---------------|---------|---------|--------|-------|
| Intercept     | -3.9900 | 1.140   | -3.500 | 0.000 |
| C(rank) [T.2] | -0.6754 | 0.316   | -2.134 | 0.033 |
| C(rank) [T.3] | -1.3402 | 0.345   | -3.881 | 0.000 |
| C(rank) [T.4] | -1.5515 | 0.418   | -3.713 | 0.000 |
| gre           | 0.0023  | 0.001   | 2.070  | 0.038 |
| gpa           | 0.8040  | 0.332   | 2.423  | 0.015 |

#### ODDS RATIOS

##### OUTPUT

|               |        |
|---------------|--------|
| Odds Ratios:  |        |
| Intercept     | 0.0185 |
| C(rank) [T.2] | 0.5089 |
| C(rank) [T.3] | 0.2618 |
| C(rank) [T.4] | 0.2119 |
| gre           | 1.0023 |
| gpa           | 2.2345 |

## INTERPRETATION:

---

- **GPA** is the strongest predictor; 1-point increase more than doubles admission odds (OR = 2.23)
- **Rank** matters significantly; Rank 1 students have 5x higher odds than Rank 4
- **GRE** has a small but significant positive effect (OR = 1.002 per point)

Problem 6

Poisson Regression GLM - Number of Awards

PROBLEM STATEMENT

The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational, general or academic) and score on their final exam in math. The data set is taken from [https://stats.idre.ucla.edu/stat/data/poisson\\_sim.csv](https://stats.idre.ucla.edu/stat/data/poisson_sim.csv) and fit the Poisson generalized linear models (GLM).

PYTHON CODE

```
1 import pandas as pd
2 import statsmodels.formula.api as smf
3
4 df = pd.read_csv('poisson_sim.csv')
5 model = smf.glm('num_awards ~ C(prog) + math', data=df,
6                 family=sm.families.Poisson()).fit()
7 print(model.summary())
```

MODEL SUMMARY

| OUTPUT        |         |         |        |       |
|---------------|---------|---------|--------|-------|
|               | coef    | std err | z      | P> z  |
| Intercept     | -5.2471 | 0.658   | -7.969 | 0.000 |
| C(prog) [T.2] | 1.0839  | 0.358   | 3.025  | 0.002 |
| C(prog) [T.3] | 0.3698  | 0.441   | 0.838  | 0.402 |
| math          | 0.0702  | 0.011   | 6.619  | 0.000 |

INCIDENT RATE RATIOS (IRR)

| OUTPUT        |        |
|---------------|--------|
| IRR:          |        |
| Intercept     | 0.0053 |
| C(prog) [T.2] | 2.9561 |
| C(prog) [T.3] | 1.4475 |
| math          | 1.0727 |

- Interpretation:
- **Math score:** 1-point increase = 7.3% more awards
  - **Program type:** Academic (baseline) vs Vocational (2.96x more awards)
  - General program not significantly different from Academic

Problem 7

Negative Binomial Regression - Days Absent

PROBLEM STATEMENT

School administrators study the attendance behavior of high school juniors at two schools. Predictors of the number of days of absence include the type of program in which the student is enrolled and a standardized test in math. The data set is taken from (<https://stats.idre.ucla.edu>)

PYTHON CODE

```
1 import pandas as pd
2 import statsmodels.formula.api as smf
3
4 df = pd.read_stata('nb_data.dta')
5 model = smf.glm('daysabs ~ C(prog) + math', data=df,
6               family=sm.families.NegativeBinomial()).fit()
7 print(model.summary())
```

MODEL SUMMARY

| OUTPUT        |         |         |        |       |
|---------------|---------|---------|--------|-------|
|               | coef    | std err | z      | P> z  |
| Intercept     | 2.6150  | 0.200   | 13.054 | 0.000 |
| C(prog) [T.2] | -0.4408 | 0.185   | -2.379 | 0.017 |
| C(prog) [T.3] | -1.2786 | 0.203   | -6.284 | 0.000 |
| math          | -0.0060 | 0.003   | -2.358 | 0.018 |

| OUTPUT                                                |  |
|-------------------------------------------------------|--|
| Mean of daysabs: 5.9554                               |  |
| Variance of daysabs: 49.5187 (Overdispersion present) |  |

- Interpretation:
- Higher math scores correlate with fewer absences
  - Academic track (baseline) has best attendance
  - General and Vocational students show significantly more absences
  - Negative Binomial appropriate due to overdispersion (variance > mean)

Problem 8

Zero-Inflated Poisson Regression - Fish Catch Data

PROBLEM STATEMENT

The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from ("<https://stats.idre.ucla.edu/stat/data/fish.csv>") and fit the Zero-Inflated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

PYTHON CODE

```
1 import pandas as pd
2 import statsmodels.api as sm
3 from statsmodels.discrete.count_model import ZeroInflatedPoisson
4
5 df = pd.read_csv('fish.csv')
6 X = df[['camper', 'persons', 'child']]
7 X = sm.add_constant(X)
8 X_infl = df[['child', 'persons']]
9 X_infl = sm.add_constant(X_infl)
10
11 model = ZeroInflatedPoisson(df['count'], X, exog_infl=X_infl).fit()
12 print(model.summary())
```

MODEL RESULTS

| OUTPUT          |         |         |         |       |
|-----------------|---------|---------|---------|-------|
|                 | coef    | std err | z       | P> z  |
| inflate_const   | 0.9813  | 0.414   | 2.370   | 0.018 |
| inflate_child   | 1.8227  | 0.315   | 5.780   | 0.000 |
| inflate_persons | -0.8454 | 0.190   | -4.456  | 0.000 |
| const           | -0.8452 | 0.166   | -5.103  | 0.000 |
| camper          | 0.7597  | 0.094   | 8.096   | 0.000 |
| persons         | 0.8337  | 0.043   | 19.587  | 0.000 |
| child           | -1.1472 | 0.091   | -12.539 | 0.000 |

## INTERPRETATION - NON-FISHER PROBABILITY (INFLATION MODEL)

### OUTPUT

Odds Ratios (Being a Non-Fisher):

|                 |        |
|-----------------|--------|
| inflate_const   | 2.6679 |
| inflate_child   | 6.1887 |
| inflate_persons | 0.4294 |

## INTERPRETATION - CATCH COUNT (COUNT MODEL)

### OUTPUT

Incident Rate Ratios (Catch Rate):

|         |        |
|---------|--------|
| const   | 0.4295 |
| camper  | 2.1377 |
| persons | 2.3019 |
| child   | 0.3175 |

## INTERPRETATION:

- **Non-fishers:** Groups with children are 6.19x more likely to be non-fishers
- **Catch rate:** Campers catch 2.14x more fish; each additional person increases catch by 130%
- **Children:** Reduce expected catch by about 68% even when fishing

# Problem 9

## Zero-Inflated Poisson - Detailed Analysis

### PROBLEM STATEMENT

The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from ("<https://stats.idre.ucla.edu/stat/data/fish.csv>") and fit the Zero-Inflated Binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

### PYTHON CODE

```
1 import pandas as pd
2 import statsmodels.api as sm
3 import statsmodels.formula.api as smf
4 import numpy as np
5
6 url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
7 fish = pd.read_csv(url)
8 fish['child'] = fish['child'].astype(int)
9
10 zip_model = sm.ZeroInflatedPoisson.from_formula(
11     "count ~ persons + child + camper",
12     fish,
13     exog_infl=fish[['persons', 'child', 'camper']],
14     inflation='logit'
15 )
16 result = zip_model.fit(method='bfgs')
17 print(result.summary())
18
19 params = result.params
20 irr = np.exp(params)
21 print(irr)
```

## MODEL OUTPUT

| OUTPUT                                 |       |         |       |       |
|----------------------------------------|-------|---------|-------|-------|
| ZeroInflatedPoisson Regression Results |       |         |       |       |
| =====                                  |       |         |       |       |
| Dep. Variable: count                   |       |         |       |       |
| No. Observations: 250                  |       |         |       |       |
| Pseudo R-squ.: 0.276                   |       |         |       |       |
| Log-Likelihood: -1032.5                |       |         |       |       |
|                                        | coef  | std err | z     | P> z  |
| inflate_persons                        | -0.45 | 0.16    | -2.81 | 0.005 |
| inflate_child                          | 0.82  | 0.20    | 4.10  | 0.000 |
| inflate_camper                         | -1.15 | 0.31    | -3.70 | 0.000 |
| Intercept                              | 1.12  | 0.15    | 7.46  | 0.000 |
| persons                                | 0.38  | 0.05    | 7.60  | 0.000 |
| child                                  | -0.56 | 0.09    | -6.22 | 0.000 |
| camper                                 | 0.71  | 0.11    | 6.45  | 0.000 |
| =====                                  |       |         |       |       |

## INTERPRETATION

### Persons

$IRR = e^{0.38} \approx 1.46$

Each additional person increases the expected number of fish caught by approximately 46%.

### Child

$IRR = e^{-0.56} \approx 0.57$

Each additional child decreases the expected fish catch by approximately 43%.

### Camper

$IRR = e^{0.71} \approx 2.03$

Campers catch about twice as many fish as non-campers.

## CONCLUSION

The ZIP model is suitable due to excess zero counts. Larger groups and campers tend to catch more fish, while the presence of children reduces fish catch.

Problem 10

Zero-Truncated Poisson - Hospital Length of Stay

PROBLEM STATEMENT

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from (["https://stats.idre.ucla.edu/stat/data/ztp.dta"](https://stats.idre.ucla.edu/stat/data/ztp.dta)) and fit the zero-truncated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

PYTHON CODE

```
1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 import statsmodels.formula.api as smf
5
6 df = pd.read_csv('ztp_temp.csv')
7 model = smf.glm('stay ~ age + hmo + died', data=df,
8                 family=sm.families.NegativeBinomial()).fit()
9 print(model.summary())
```

MODEL RESULTS

| OUTPUT    |         |         |        |       |
|-----------|---------|---------|--------|-------|
|           | coef    | std err | z      | P> z  |
| Intercept | 2.4378  | 0.091   | 26.920 | 0.000 |
| age       | -0.0147 | 0.016   | -0.894 | 0.371 |
| hmo       | -0.1375 | 0.075   | -1.845 | 0.065 |
| died      | -0.2038 | 0.058   | -3.505 | 0.000 |

## INCIDENT RATE RATIOS

### OUTPUT

|           |         |
|-----------|---------|
| IRR:      |         |
| Intercept | 11.4473 |
| age       | 0.9854  |
| hmo       | 0.8715  |
| died      | 0.8156  |

### INTERPRETATION:

- **Died in hospital:** 18.4% shorter stays (p < 0.001)
- **HMO insurance:** 12.8% shorter stays (marginally significant)
- **Age:** Not statistically significant

Problem 11

Zero-Truncated Negative Binomial - Hospital Length of Stay

PROBLEM STATEMENT

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from (<https://stats.idre.ucla.edu/stat/data/ztp.dta>) and fit the zero-truncated negative binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

PYTHON CODE

```
1 from statsmodels.base.model import GenericLikelihoodModel
2 import numpy as np
3
4 class ZTNB(GenericLikelihoodModel):
5     def loglike(self, params):
6         beta = params[:-1]
7         alpha = params[-1]
8         if alpha <= 0: return -np.inf
9         mu = np.exp(np.dot(self.exog, beta))
10        size = 1 / alpha
11        prob = size / (size + mu)
12        ll_nb = nbinom.logpmf(self.endog, size, prob)
13        p0 = (1 + alpha * mu) ** (-1/alpha)
14        return np.sum(ll_nb - np.log(1 - p0))
15
16 model = ZTNB(df['stay'], X).fit()
```

MODEL RESULTS

| OUTPUT |         |         |        |       |
|--------|---------|---------|--------|-------|
|        | coef    | std err | z      | P> z  |
| const  | 2.4083  | 0.072   | 33.457 | 0.000 |
| age    | -0.0157 | 0.013   | -1.198 | 0.231 |
| hmo    | -0.1471 | 0.059   | -2.484 | 0.013 |
| died   | -0.2177 | 0.046   | -4.717 | 0.000 |
| alpha  | 0.5663  | 0.031   | 18.132 | 0.000 |

## INCIDENT RATE RATIOS

### OUTPUT

|       |         |
|-------|---------|
| IRR:  |         |
| const | 11.1152 |
| age   | 0.9844  |
| hmo   | 0.8632  |
| died  | 0.8043  |

## FINAL INTERPRETATION:

- **Significant predictors:** Mortality and HMO insurance
- Patients who die have 19.6% shorter stays
- HMO patients have 13.7% shorter stays than privately insured
- Age does not significantly impact length of stay
- Zero-truncated negative binomial model appropriate due to:
  - Minimum stay of 1 day (zero truncation)
  - Overdispersion in the data (alpha significantly  $\neq 0$ )